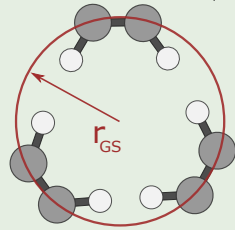


Quantifying framework flexibility

Define a CV: $s = S(\mathbf{R})$



Structure generation

MD sampling of framework flexibility

Group by collective variable

Bin structures by flexible CV: s
Create set of N_s geometries $\{\mathbf{R}\}_s$

Constrained optimized structures

Perform constrained minimizations
Range of geometries: $\mathbf{R}_0(s)$

Approach 1

Representative geometry

Single geometry per s
One cDFT calc. $\rightarrow \rho(\mathbf{r}|\mu, s)$

Approach 2

Effective potential

Boltzmann avg. potential:

$$v_{\text{eff}}(\mathbf{r}, s) = -\frac{1}{\beta} \ln \left[\frac{1}{N_s} \sum_{\mathbf{R}_i} e^{-\beta v(\mathbf{r}, \mathbf{R}_i)} \right]$$

One cDFT calc. $\rightarrow \rho(\mathbf{r}|\mu, s)$

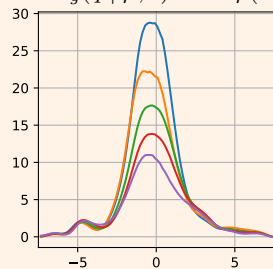
Approach 3

Density averaging

N_s cDFT calcs. $\rightarrow \{\rho(\mathbf{r})\}_s$
Average densities:
 $\rho(\mathbf{r}|\mu, s) \approx \frac{1}{N_s} \sum_{\mathbf{R}_i} \rho(\mathbf{r}|\mu, \mathbf{R}_i)$

Free energy profiles

Get FEP $\Omega_g(q | \mu, s)$ from $\rho(\mathbf{r} | \mu, s)$



Flexibility integration

Thermodynamic min. over $F_h(s)$
 $\Omega(q|\mu) = \min_s [F_h(s) + \Omega_g(q | \mu, s)]$

Flexible diffusion profile

Extract diffusivity through TST

Empty host FEP vs. s

